JATIN PATHAK

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PROFESSIONAL SUMMARY

Electronics & Telecommunication engineering student with hands-on experience in backend, full-stack, AI/GenAI, and cloud development. Skilled in Node.js, Express.js, React, and MongoDB, with practical exposure to LLMs, RAG, and AWS. Interested in building secure, scalable applications and applying software, AI, and cloud technologies to real-world problems.

PROFESSIONAL EXPERIENCE

Bluestock Fintech **2 Months**
Backend Developer Intern

- Developed robust backend services using Node.js, Express.js, and MongoDB, successfully implementing user authentication, contact form submissions, and early-access registration flows, contributing to a seamless user experience.
- Engineered and deployed 5+ RESTful API endpoints, enhancing application communication for authentication, contact forms, and subscription management, ensuring smooth data flow.
- Implemented comprehensive input validation, error handling, and duplicate email checks, resulting in reliable API operations and maintaining high data integrity for the platform.
- Tested and debugged APIs using Postman, actively managed code versions with Git/GitHub to ensure code quality and facilitate efficient development and track changes.

TECHNICAL SKILLS

Languages: C++, TypeScript, JavaScript, Python (Basics)
AI / GenAI: LLMs, RAG, LangChain, Prompt Engineering, Tool Calling
Web Development: HTML, CSS, Tailwind CSS, React (Intermediate), Node.js, Express.js
Databases: PostgreSQL, MongoDB, MySQL (Basics)
Cloud / DevOps: AWS (EC2, S3, IAM), Docker, GitHub Actions
Developer Tools: Cursor, Antigravity, Git, GitHub, Postman, VS Code
Core Concepts: REST APIs, DBMS, Computer Networks

PROJECTS

Deep Packet Inspection (DPI) Firewall **GitHub**
Node.js, TypeScript, Wireshark, Worker Threads

- Developed a DPI firewall using Node.js and TypeScript to identify applications and filter network traffic.
- Implemented packet parsing, 5-tuple tracking, and TLS SNI inspection to identify applications and filter domains.
- Built a multi-threaded packet processing system using Worker Threads to track network sessions and improve filtering efficiency.

Vault Backend – Money Transfer System **GitHub**
Node.js, Express.js, MongoDB, JWT, Nodemailer

- Built a money transfer backend using Node.js, Express.js, and MongoDB for account management and fund transfers.
- Implemented a 10-step transaction workflow with balance checks, duplicate prevention, and automatic rollback.
- Secured APIs with JWT, bcryptjs, and MongoDB transactions, and automated receipt emails using Nodemailer.

CipherShield – Cybersecurity Platform **GitHub | LIVE**
React, Node.js, Express, MongoDB, REST APIs

- Built a cybersecurity platform with a React admin dashboard for secure text and file encryption.
- Integrated JWT authentication to ensure secure user access and data protection.
- Developed a cryptographic engine using NPCR and UACI metrics to evaluate and benchmark encryption security.

EDUCATION

THAKUR COLLEGE OF ENGINEERING AND TECHNOLOGY **2023–2027**
B.E. in Electronics & Telecommunication Mumbai, India

CERTIFICATIONS & EXTRACURRICULAR

- JavaScript Bootcamp – LetsUpgrade (LUEJSSEP124466)
- **Academic Tutor**
  Mentored 25+ students as an academic tutor, enhancing communication and interpersonal abilities.